

Problem

Find the Laplace transform of the following functions:

(a) $t \cos tu(t)$

(b) $e^{-t} \sin tu(t)$

(c) $\frac{\sin \beta t}{t} u(t)$

Step-by-step solution

Step 1 of 2

(a) $f(t) = t \cos t u(t)$

$$L[\cos t u(t)] = \frac{s}{s^2 + 1}$$

$$\begin{aligned} L[t \cos t u(t)] &= \frac{-d}{ds} \left[\frac{s}{s^2 + 1} \right] \\ &= \left[\frac{s^2 + 1 - s(2s)}{(s^2 + 1)^2} \right] \\ &= \boxed{\frac{s^2 - 1}{(s^2 + 1)^2}} \end{aligned}$$

(b) $e^{-t} t \sin t u(t)$

$$L[\sin t u(t)] = \frac{1}{s^2 + 1}$$

$$\begin{aligned} L[t \sin t u(t)] &= \frac{-d}{ds} \left[(s^2 + 1)^{-1} \right] \\ &= - \left[-(s^2 + 1)^{-2} 2(s) \right] = \frac{2s}{(s^2 + 1)^2} \end{aligned}$$

$$L[e^{-t} t \sin t u(t)] = \boxed{\frac{2(s+1)}{[(s+1)^2 + 1]^2}}$$

Step 2 of 2

$$(c) \frac{\sin \beta t}{t} u(t)$$

$$L[\sin \beta t \cdot u(t)] = \frac{\beta}{s^2 + \beta^2}$$

By frequency integration.

$$\begin{aligned} L\left[\frac{\sin \beta t}{t} u(t)\right] &= \int_s^\infty \frac{\beta}{s^2 + \beta^2} ds \\ &= \beta \left[\frac{1}{\beta} \tan^{-1}\left(\frac{s}{\beta}\right) \right]_s^\infty \\ &= \tan^{-1}(\infty) - \tan^{-1}\left(\frac{s}{\beta}\right) \\ &= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{\beta}\right) = \cos^{-1}\left(\frac{s}{\beta}\right) \\ &= \boxed{\tan^{-1}\left(\frac{\beta}{s}\right)} \end{aligned}$$